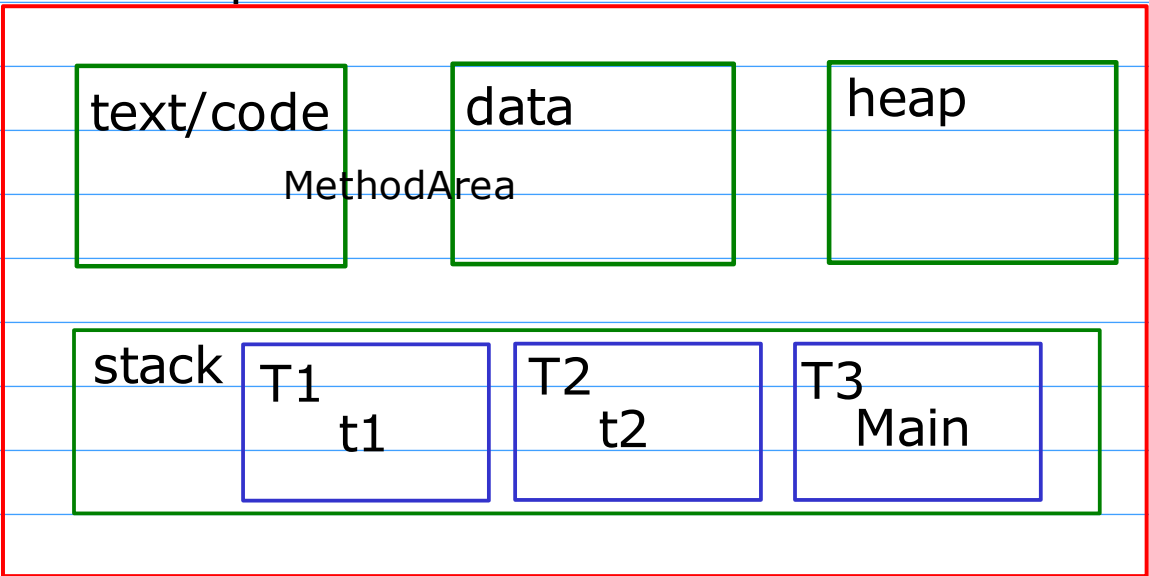


MultiThreading

RAM -> JVM(Process) ->

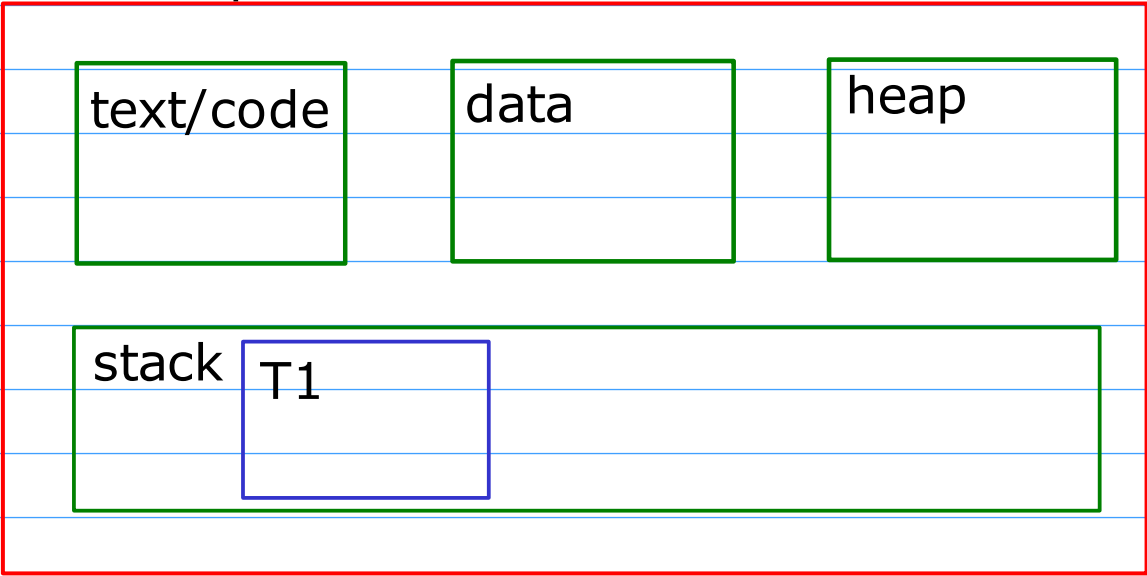
Process p1



int main(){

T1	for(1 to 10) sleep(1s)	✓	0s + 10s
T2	for(11 to 20) sleep(1s)	✓	10s + 10s
T3	for(21 to 30) sleep(1s)	✓	20s + 10s
}			

Process p2

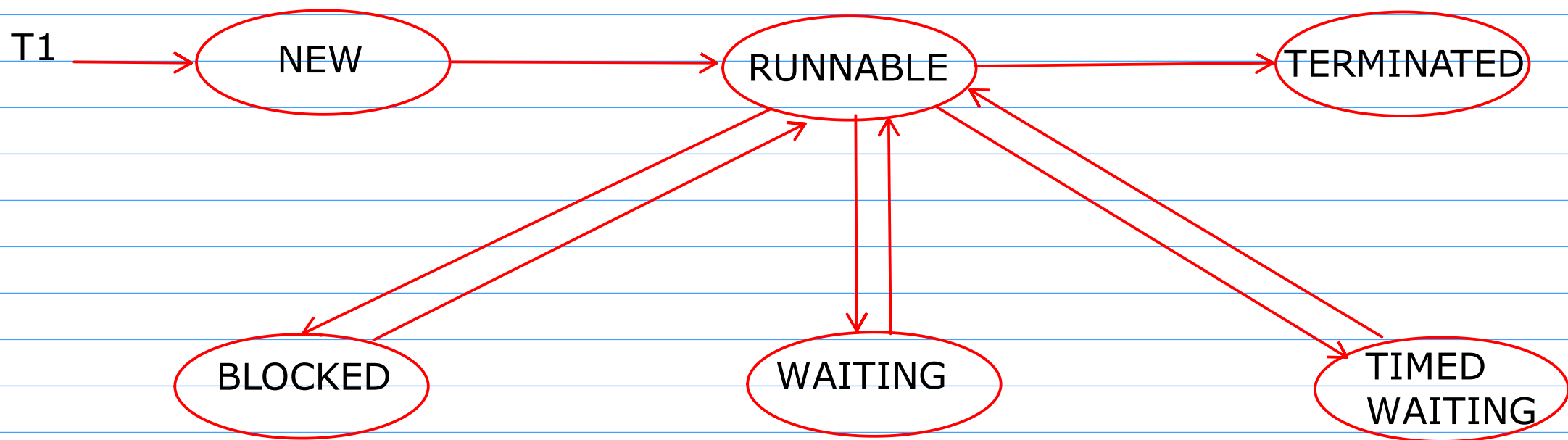


10s

- # In java Thread can be created in 2 ways
- 1. By creating a subclass of Thread class
If a class do not have any superclass then we can use the Thread class as super class to create a thread
 - 2. By creating a subclass if Runnable interface
- In java we cannot have multiple class inheritance, hence Runnable interface helps in creating thread for the classes which already have a Superclass

java Program

- Main Thread
main()
 - MyThread
run()
 - for loop 7s
 - MyRunnable
run()
 - for loop 10s
 - for loop 5s
 - sysout("Program finished")



Ideal Case

Deposit

Withdraw

double newBalance = balance + 10000;
balance = newBalance;

double newBalance = balance - 10000;
balance = newBalance;

BankAccount

20000

newBalance

~~10000~~
~~20000~~
10000

balance

10000

newBalance

Race Condition

Deposit

Withdraw

double newBalance = balance + 10000;
balance = newBalance;

double newBalance = balance - 10000;
balance = newBalance;

BankAccount

20000

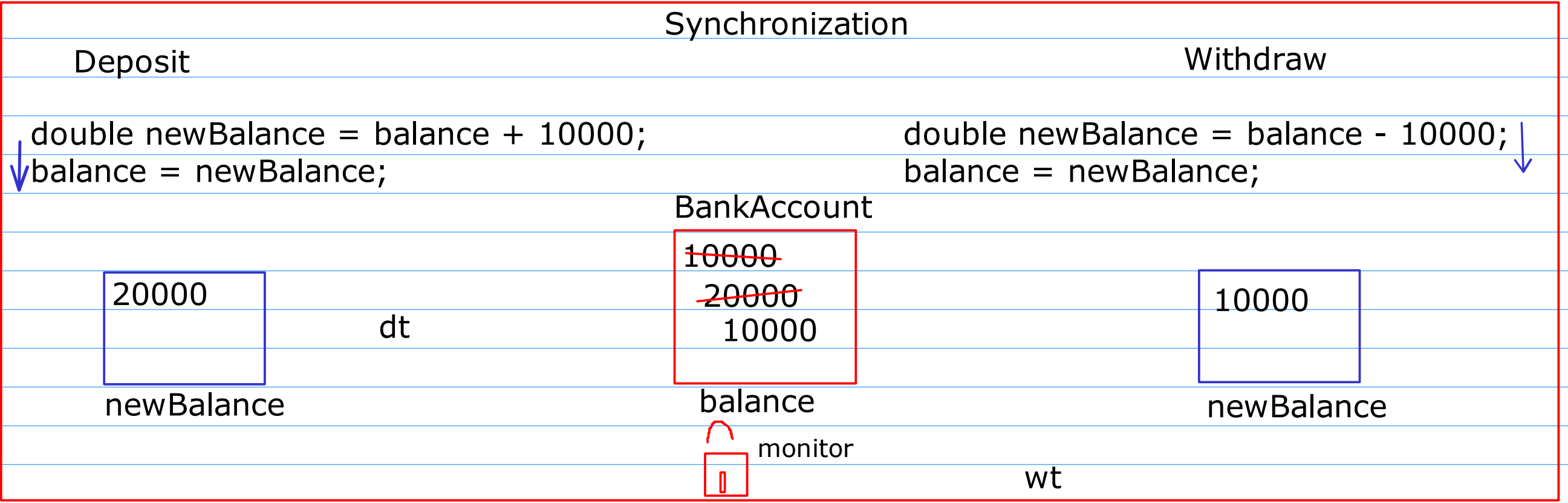
newBalance

~~10000~~
~~0~~
20000

balance

0

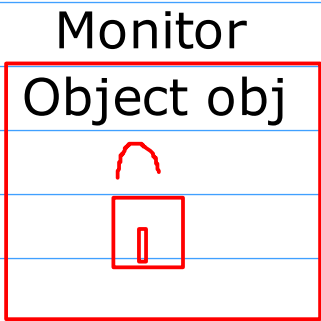
newBalance



Sunbeam

```
for(){  
  
}
```

st



it

Infotech

```
for(){  
  
}
```

20*4 = 80 hrs

OOP

Language Fundamentals

Class Object

Basics of Java

static, hirerachy

Features given by Java

- Exception Handling
- Generics
- Collection Framework

Functional Programming

- Java 8 interface
- Functional interface
- Lamda Expression

Stream Programming

IO FrameWork

Reflection and Annotation

MultiThreading